

Networking Questions & Answers

1. Which of the following messages in the DHCP process are broadcasted? (Choose two)

Correct Answers:

- C. Discover
- A. Request

Explanation:

- DHCP Discover: Sent as a broadcast by the client to find available DHCP servers.
- DHCP Request: Also broadcasted to inform all DHCP servers of the chosen offer.

2. Which command would you use to ensure that an ACL does not block web-based TCP traffic?

Correct Answer:

- B. permit tcp any any eq 80

Explanation:

- This permits TCP traffic from any source to any destination on port 80 (HTTP).

3. Explain Network Topologies

Network topology refers to the physical or logical layout of a network. Types include:

- Bus: Single central cable.
- Star: Devices connected to a central hub.
- Ring: Circular connection.
- Mesh: Interconnected devices.
- Hybrid: Combination of types.

4. Explain TCP/IP Networking Model

The TCP/IP Model has 4 layers:

1. Application (e.g., HTTP, FTP)
2. Transport (TCP/UDP)
3. Internet (IP)

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4. Network Access (Ethernet, Wi-Fi)

5. Explain LAN and WAN Network

- LAN: Local Area Network, small area, high speed.
- WAN: Wide Area Network, large area (e.g., Internet), slower.

6. Explain Operation of Switch

- Operates at Layer 2 (Data Link).
- Learns MAC addresses to forward frames.
- Creates separate collision domains.
- In full duplex mode, eliminates collisions.

7. Describe the Purpose and Functions of Various Network Devices

Router: Connects networks.

Switch: Connects LAN devices.

Hub: Broadcasts data.

Access Point: Wireless extension.

Modem: Internet access.

Firewall: Filters traffic.

8. List of Appropriate Media, Cables, Ports, and Connectors to Connect Switches

Ethernet: Cat5e/Cat6, RJ-45, Ethernet port

Fiber Optic: Single/Multi-mode, LC/SC, SFP port

Console: Rollover cable, RJ-45 to DB-9/USB, Console port

9. Define Network Devices and Hosts

- Network Devices: Hardware that routes or manages traffic (routers, switches).
- Hosts: End devices like PCs, phones, printers.